

Data Science

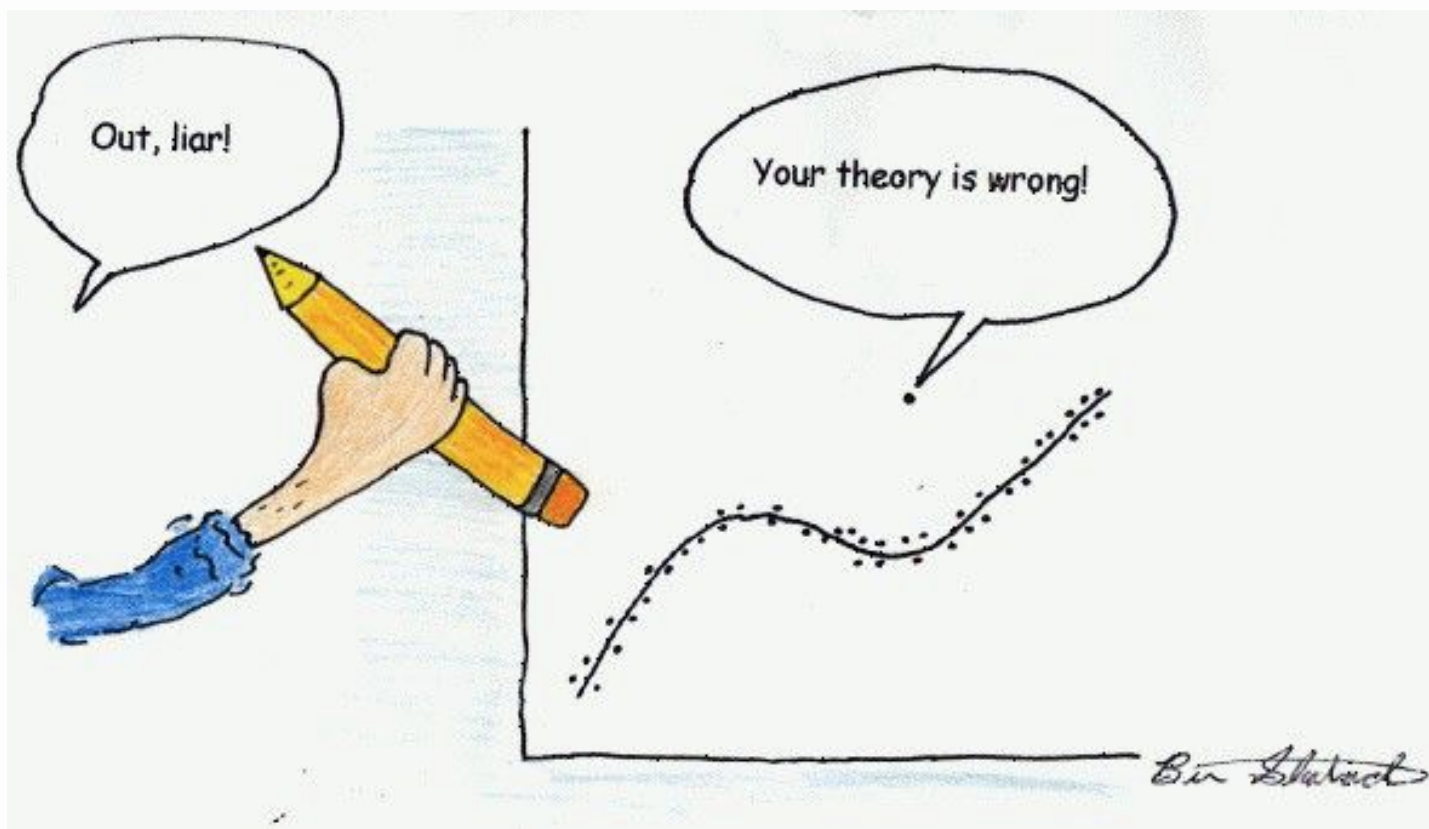
Outliers

Prof. Rayson Laroca
rayson@ppgia.pucpr.br

Peer Review

- Right after class, I will assign two projects to each team for the peer review of the checkpoint
 - The task instructions and template are already available on Canvas
 - The specific teams your group will be assigned to evaluate will be sent via a comment on the task
- I will compare your team's reviews with my own evaluation

OUTLIERS



Learning Objectives

- By the end of this class, you should be able to:
 - explain what an outlier is
 - identify possible causes of outliers
 - detect potential outliers using box plots and Tukey's rule
 - understand the intuition behind Isolation Forest
 - decide whether an outlier should be investigated, corrected, kept, or removed

What is an outlier?

- An outlier is a data point that is unusually distant from the others
- It may be:
 - Much larger or much smaller than most observations
 - Inconsistent with the general pattern of the data
- Outliers are **context-dependent**
 - A value that is unusual in one dataset may be normal in another

Why do outliers appear?

- Measurement or data acquisition errors
- Data entry or preprocessing mistakes
- Corrupted or missing information encoded incorrectly
- Natural rare events
- Legitimate but extreme observations
- **Takeaway**: not every outlier should be removed!

Example 1

- Which value looks suspicious?

Gender	Age	Height
M	20	159
F	21	191
M	24	173
M	24	181
F	28	156
M	26	192
F	19	280
F	22	162
M	26	190

Example 1

- Which value looks suspicious?
 - Height = 280 is suspicious
 - But before removing it, we should verify whether:
 - the unit is correct
 - the value was typed incorrectly
 - the observation is real

Gender	Age	Height
M	20	159
F	21	191
M	24	173
M	24	181
F	28	156
M	26	192
F	19	280
F	22	162
M	26	190

Example 2

- And here?

Athlete	# of medals
Michael Phelps	28
Larisa Latynina	18
Marit Bjorgen	15
Nikolai Andrianov	15
Ole Einar Bjorndalen	13
Boris Shakhlin	13
Edoardo Mangiarotti	13

Example 2

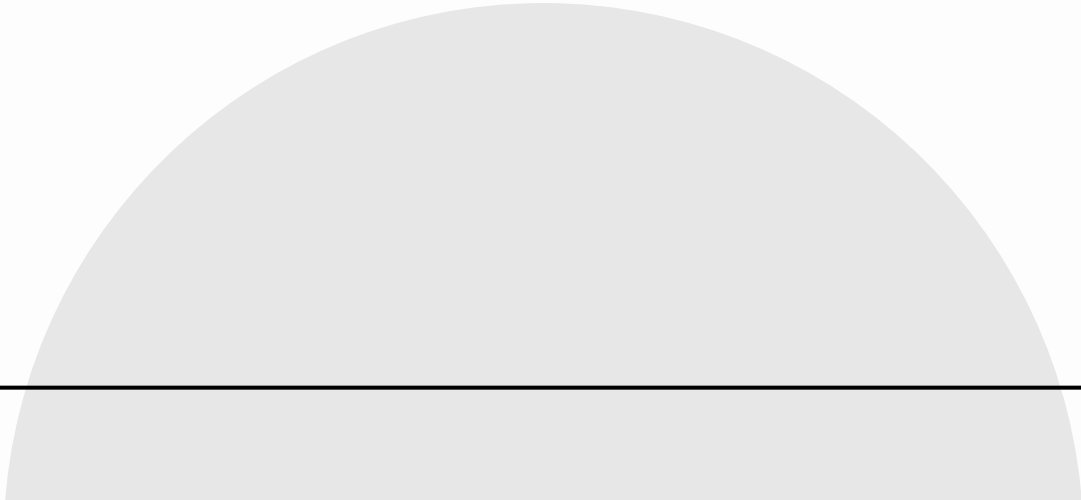
- And here?
 - Statistical rarity does not imply data error
 - **Domain knowledge matters**

Athlete	# of medals
Michael Phelps	28
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After Detecting an Outlier

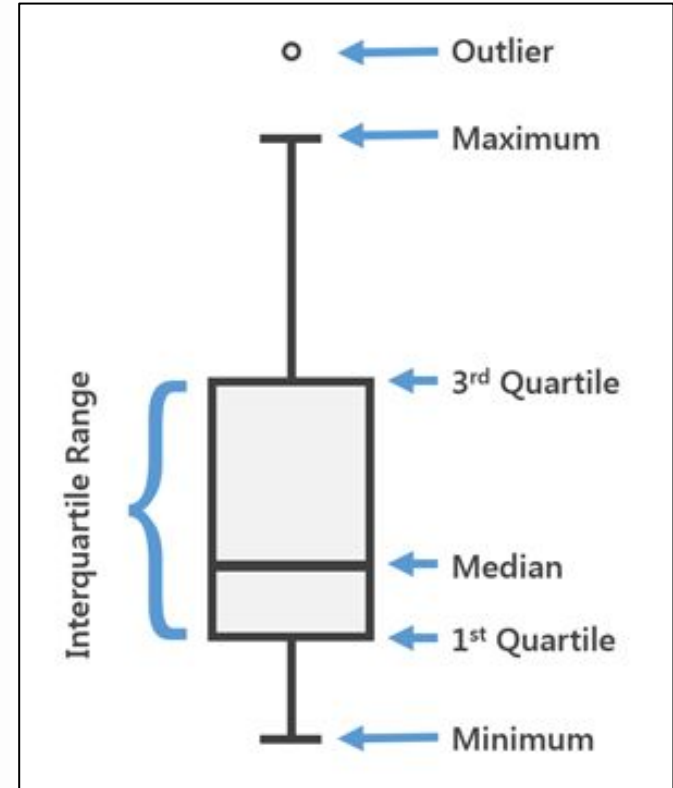
- Possible actions include:
 - verify the original data source
 - correct the value if it was entered incorrectly
 - keep the observation if it is valid
 - remove the observation if it is clearly an error
 - analyze it separately if it represents a rare but important case
- **Rule of thumb**
 - investigate first
 - remove later

OUTLIER DETECTION WITH BOX PLOTS



Box Plots – Recap

- A box plot summarizes a distribution using **quartiles**
- Key components:
 - **Q1 (1st quartile)** → 25% of the data
 - **Q2 (Median)** → 50% of the data
 - **Q3 (3rd quartile)** → 75% of the data
 - **Interquartile Range (IQR):** $Q3 - Q1$
 - **Maximum** = $Q3 + 1.5 * IQR$
 - **Minimum** = $Q1 - 1.5 * IQR$



Tukey's method

- Tukey's method is based on quartiles, more specifically on the interquartile range (IQR)
- General idea:
 - Apply Tukey's method to each variable individually
 - If an instance is identified as an outlier n times, we will analyze for removal
 - **How do we define n ?** Usually $n > 1$ facilitates our analysis

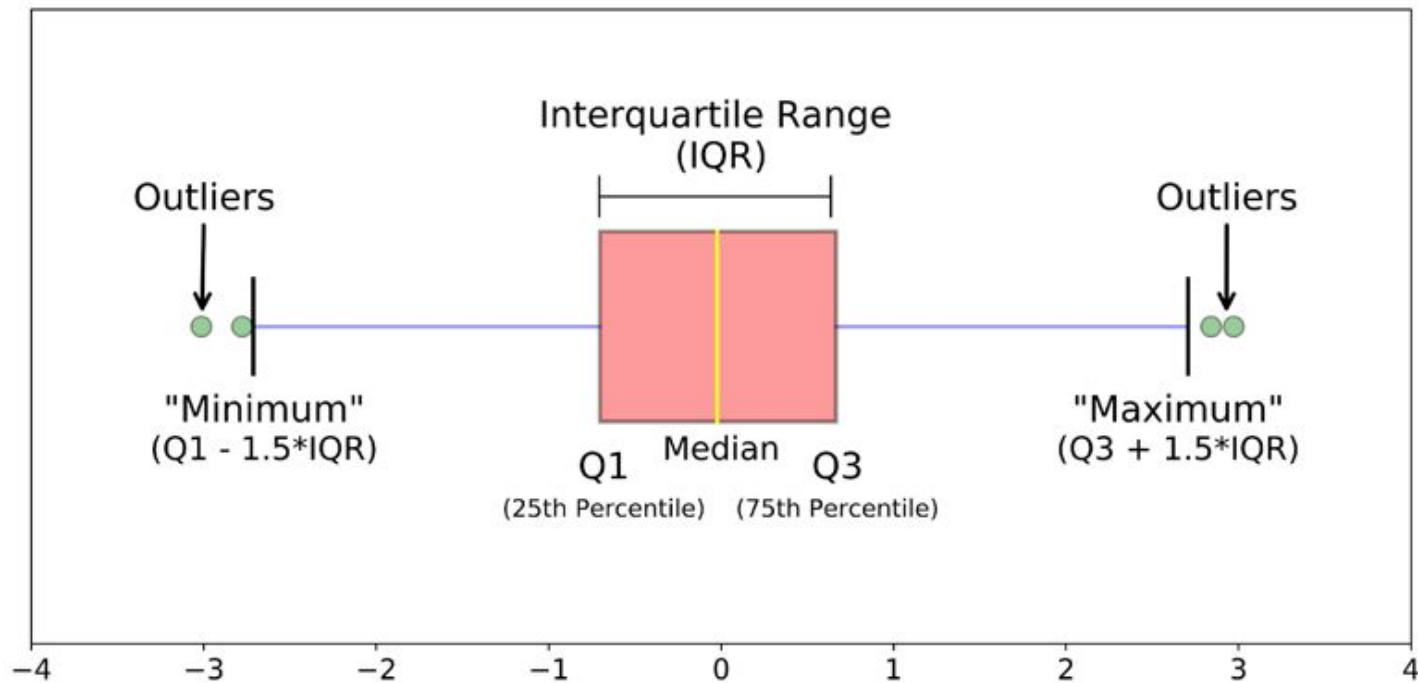
Interquartile range (IQR)

- Q1: 1st quartile (25th percentile)
- Q3: 3rd quartile (75th percentile)
- **IQR** = $Q3 - Q1$
 - The IQR measures the spread of the middle 50% of the data

IQR rule

- A point is considered an outlier if:
 - $x < Q1 - 1.5 \times IQR$
 - or
 - $x > Q3 + 1.5 \times IQR$
- Common thresholds:
 - **k = 1.5** → standard rule of thumb
 - **k = 3** → extreme outliers
 - **Note:** these are heuristic thresholds, not absolute truth

Box plot



Why Median and IQR?

- The mean and the standard deviation are sensitive to extreme values
 - The median and the IQR are more robust
- This is why Tukey's method is often a good starting point for outlier detection
- Robust statistics help us describe the data without being overly influenced by a few extreme observations

Tukey's Method: Steps

1. Sort the values from smallest to largest
2. Find Q1 and Q3
 - a. **Approach 1:** exact-position check
 - i. Compute 25% of n and 75% of n
 - ii. If each result matches exactly one position in the ordered list, use those values
 - iii. Example: if $n = 12$, then Q1 is the 3rd value and Q3 is the 9th value

Tukey's Method: Steps

1. Sort the values from smallest to largest
2. Find Q1 and Q3
 - a. **Approach 2:** median of the halves
 - i. Find Q2 (the median)
 - ii. Split the ordered data into a lower half and an upper half
 - iii. Q1 is the median of the lower half
 - iv. Q3 is the median of the upper half
 - v. If n is odd, do not include Q2 in either half

Tukey's Method: Steps

1. Sort the values from smallest to largest
2. Find Q1 and Q3 using either approach
3. Compute the IQR
 - a. $IQR = Q3 - Q1$
4. Find the limits
 - a. Lower limit = $Q1 - 1.5 \times IQR$
 - b. Upper limit = $Q3 + 1.5 \times IQR$
5. Identify outliers
 - a. Any value below the lower limit or above the upper limit

PRACTICE

Enron

- Enron was once one of the largest companies in the United States
- In 2001/2002, it collapsed due to major accounting fraud
- Part of the company data became public
 - We will explore this dataset to conduct outliers analysis



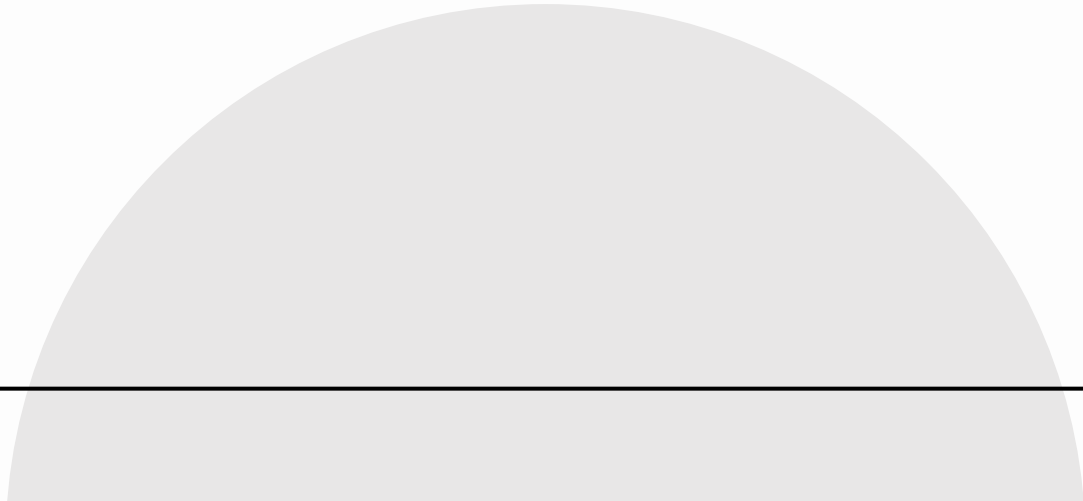
Practice

- Let's investigate outliers in practice!
 - `df =`
`pd.read_csv('https://raysonlaroca.github.io/teaching/data-science/datasets/enron.csv')`

Limitations of Univariate Detection

- Tukey's rule is usually applied one variable at a time
 - This works well for univariate outliers
- However, some observations may look normal in each variable separately
 - Still, they may be unusual when we consider the combination of variables. For example:
 - a person may have a normal height
 - and a normal weight
 - but the combination of both may still look inconsistent

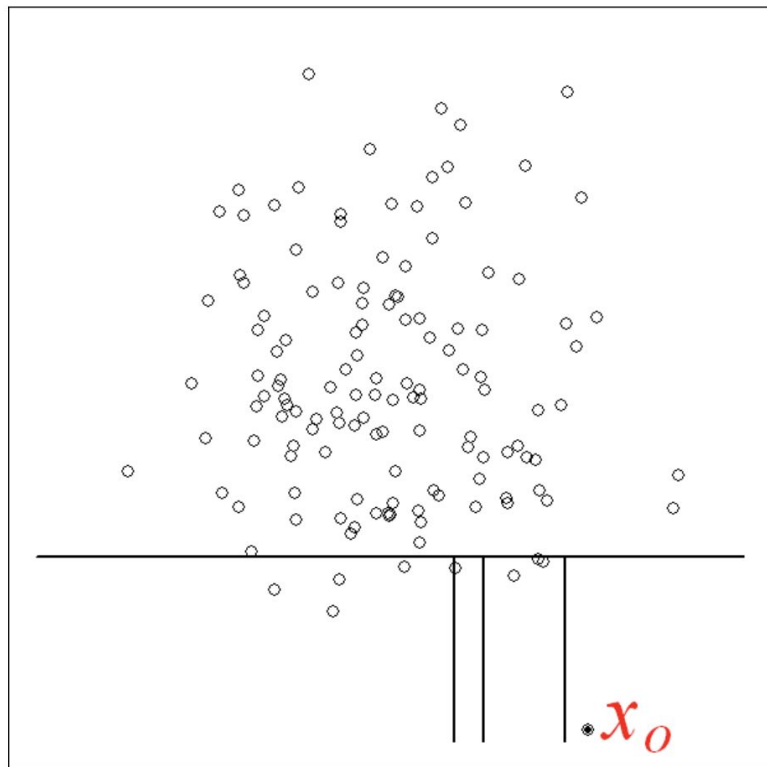
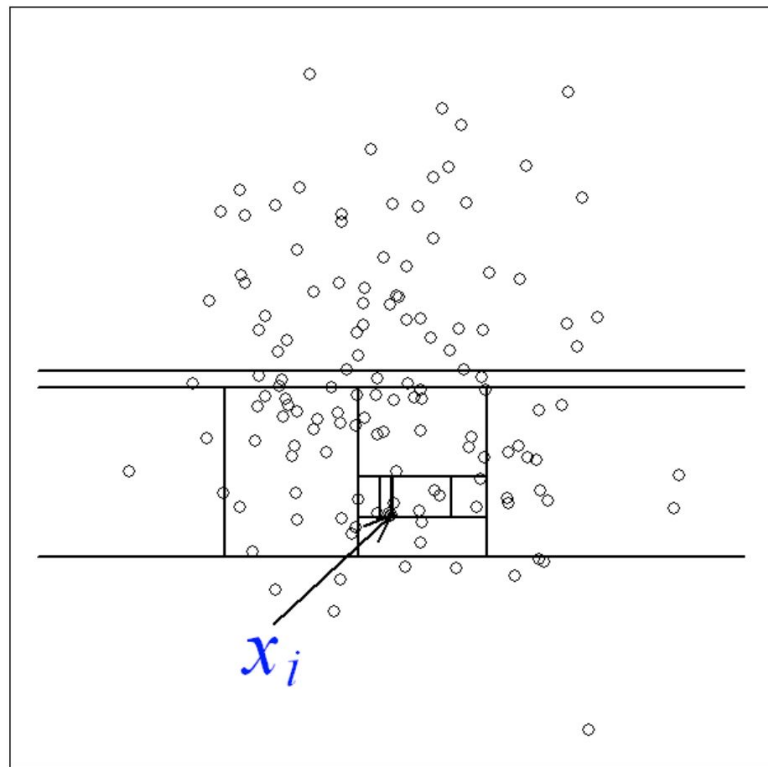
ISOLATION FOREST



Isolation Forest

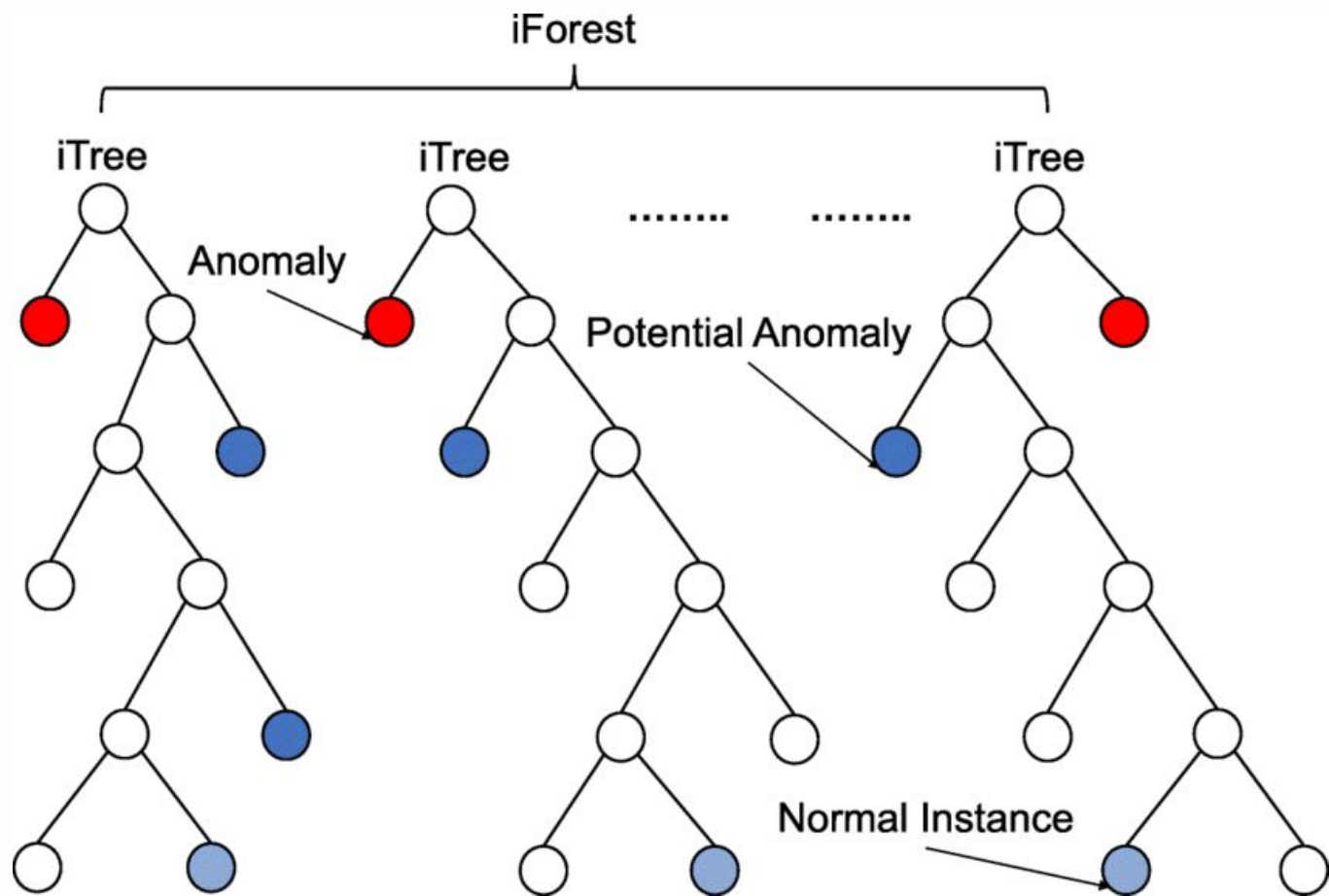
- Another popular approach for anomaly/outlier detection
- Main idea:
 - Anomalies are rare
 - Anomalies are different from normal observations
 - Therefore, they are easier to isolate
- Normal points usually require more splits to be separated
vs.
- Outliers usually require fewer splits

Visual Intuition



How does Isolation Forest work?

- A forest of trees, each called an **isolation tree** (iTree).
- Each tree is built by repeatedly:
 - Randomly selecting a feature
 - Randomly selecting a split value within that feature range
 - Partitioning the data until:
 - a point becomes isolated, or
 - the maximum tree depth is reached
- **Key idea:** fewer splits to isolate a point usually indicate higher anomaly likelihood



Practice with Isolation Forest

- Let's employ isolation forest in practice!
 - `df =`
`pd.read_csv('https://raysonlaroca.github.io/teaching/data-science/datasets/enron.csv')`

Wrap up

- Outliers can be errors, rare events, or valid extreme cases
 - Outlier detection should always be combined with data inspection and domain knowledge
- Tukey's method provide a simple and interpretable starting point
 - Usually applied to one variable at a time
- Isolation Forest can detect anomalies in a more flexible way
 - Useful in multivariate settings, but less interpretable than a box plot

Next steps

- Continue working on your project
 - Investigate outliers in your own dataset
- Complete the checkpoint peer review by Tuesday, April 21
- Ask any questions you have about next week's exam